DRAFT, not for distribution. Readable version generated from the LaTeX sources (tools/tex_to_html.py); the layout differs from the VJCS template.

When Rules Should Decide: Answering, Clarifying and Refusing Real Citizen Questions about Commune-Level Administrative Procedures in Vietnam

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Abstract. An assistant for the procedures of commune and ward offices must decide whether to answer, clarify, or refuse. We study it on real citizen questions from two Vietnamese public-service portals, with locked held-out sets and one run per configuration; a second label pass on 100 questions gives κ=0.685. On one frozen system, behavior accuracy falls from 77.2% on team-written questions to 56.5% on real ones, 79% of which must be refused. A transparent rule layer raises balanced accuracy from 0.647 to 0.814 and end-to-end answers from 19 to 35 of 59, with no net change on the team-written set; on a second locked set it raises correct refusals from 175 to 297 of 335. A reranker, a local 4-billion-parameter LLM and a second round of rules add no significant end-to-end gain, and a retrieval-augmented baseline where the model decides scores below always refusing.
Keywords: Administrative procedures; question answering; abstention; Vietnamese; rule-based systems; evaluation.

1. Introduction

Many everyday administrative procedures in Vietnam are handled by commune-level offices: civil status, residence, identity cards, certified copies, social benefits. The National Public Service Portal publishes the official record of each procedure[1], but citizens rarely ask in its language. They write letters, with a greeting, a family story, abbreviations, often no diacritics, and somewhere a question. Many questions are not about carrying out a procedure at all, but about a file already submitted, a complaint, a fine, or the benefits of police officers. Answering these from procedure records gives confident but wrong guidance, which costs citizens time and trust[2,3].

The task is therefore a three-way decision before it is a retrieval problem: *answer* with one procedure, *clarify* when several fit, or *refuse*. Abstention is a known weak point of language models[4,5], and intent classifiers struggle with out-of-scope queries[6], yet prior work on Vietnamese public-service question answering evaluates only answerable questions[7].

We study this decision on real citizen questions, for an assistant that covers only procedures handled at the commune level (communes, wards and special zones), not all administrative procedures in Vietnam, and that runs on one local machine so that questions need not leave the office[8]. We report four findings.

- **Our team-written test set overstated performance**: 79% of real questions are out of scope (88–92% in random samples), against 11% of the team's, and top-1 accuracy falls from 71.9% to 33.9%.
- **A transparent rule layer closes much of the gap** on two locked test sets, mostly by refusing correctly, with no net change on the team-written set.

- **Learned components do not replace the rules**: a reranker and a local 4B LLM add no significant end-to-end gain, and letting the model decide scores below always refusing.
- **Development gains need a fresh test**: a second round of rules that looked better on development ties the first on new questions.

Throughout, development and test sets are separated, each test set is locked by hash and commit before the code it evaluates, each configuration runs once on frozen code, and every configuration is reported, including those that lost.

2. Related Work

Vietnamese public-service and legal question answering. Pham Duy and Le Thanh retrieve legal passages for public-service questions, using 4,547 question–answer pairs from the portal's frequently-asked-questions pages and 785,996 passages, with BM25+, a contrastively trained PhoBERT encoder and a cross-encoder, evaluated by Recall@k, MAP and NDCG[7]. ViGPTQA trains Vietnamese instruction-following LLMs and evaluates them on question answering, including law[9]; Hien et al. combine vector retrieval over 45,000 legal documents with an LLM, evaluated with RAGAS and a user study[10]; their system routes out-of-scope questions to experts, but this step is not evaluated. None of these measures refusal on real citizen questions.

Abstention and out-of-scope detection. Adding unanswerable questions to SQuAD caused a large drop for strong readers[11]; a trained calibrator helps a reader abstain under domain shift[5]; on CLINC150, intent classifiers that do well in scope struggle to flag out-of-scope queries[6]. A survey frames abstention from the query, the model and human values[4], and clarifying questions are a third option between answering and refusing[12].

Who decides. Retrieval-augmented generation conditions a generator on retrieved passages[13]; Self-RAG lets the model decide when to retrieve and critique its output[14], and corrective RAG lets a retrieval evaluator choose an action[15]. Programmable guardrails take the opposite route: developer-written rails, independent of the LLM, constrain the dialogue[16]. In e-government, a proposed governance pipeline adds validation, monitoring and escalation to humans[2], and a benchmark for governmental use finds that answering correctly does not imply acknowledging what one does not know[3]. We compare rules, a reranker score and a local LLM as the decider under one protocol.

3. Task and Data

Task. For a citizen's question the system outputs *answer* with a procedure identifier, *clarify* with candidate procedures, or *refuse* with a short reason. A question is labeled *answer* when exactly one commune-level procedure (or a pair of equivalent ones) is clearly what the citizen needs; *clarify* when several different procedures fit, or when the procedure exists only in province-specific versions and no province is given; *refuse* when the procedure is not in the data or not handled at commune level, or the question concerns the status of a submitted file, a complaint or petition, law and penalties, police internal matters, or no procedure at all.

Procedure data. The National Public Service Portal[1] was collected once on 13 October 2026: 1,517 procedures, including province-specific versions for 34 provinces. The 542 commune-level procedures that are not province-specific form the scope. Records hold name, field, agency, documents, fees, processing time, steps and legal basis,

and no personal data. The portal has no official API; we used its public, unauthenticated JSON endpoints at two requests per second on one thread.

Real questions. We pooled two public Hugging Face datasets of citizen questions sent to the Binh Dinh provincial portal[17] and the Ministry of Public Security portal[18]. Personal data were masked before labeling. After removing duplicates, lines under four words, greetings, and 266 questions used earlier to design clarification rules, the pool has 1,490 questions. Samples were drawn with fixed seeds from a *random* stratum and a *topical* stratum (keywords of commune-level fields, written before any system output was seen, to obtain more answerable questions):

- **Development set**: 282 questions after removing non-questions (55 answer, 7 clarify, 220 refuse).
- **First test set**: 313 other questions (59 answer, 6 clarify, 248 refuse), locked by SHA-256 and a commit before any code change. Eleven of them are near-duplicates of development questions (resent by the same citizens); results without them are given in Section 6.2.
- **Second test set**: the rest of the pool, random stratum only, in an order locked before the second improvement round. An author labeled the first 400, with labels proposed by Gemini 4 Argon and reviewed item by item; 363 remain after removing non-questions (21 answer, 7 clarify, 335 refuse). The AI assistant that wrote the rules (Section 4) never read these questions or labels before the final run.

Labels and agreement. Labels for the development and first test sets were produced, following a written guideline, with the same AI assistant that wrote the rules (for the development set, two independent passes, reconciled and reviewed by the project lead). A random sample of 100 first-test questions was labeled again: Gemini 4 Argon proposed labels and an author checked every item without seeing the first pass, although that author had seen the aggregate class proportions. The second pass marked 9 petitions as non-questions; on the other 91 the passes agree on 79 (86.8%), Cohen's κ=0.685[19] (95% bootstrap interval 0.55–0.82), substantial agreement[20]. Most disagreements are *answer* versus *clarify* between closely related procedures. The first-pass labels were not changed.

Team-written set. For comparison we use the blind part of the team's regression set: 518 questions written over several rounds, 102 of them imitating citizen phrasing, with 384 *answer* (381 whose procedure still exists), 76 *clarify* and 58 *refuse*.

4. System and Variants

Base system. The assistant as it stood before this study uses no language model to decide. A question is split into segments (one per requested procedure), each normalized (abbreviations, synonyms, the fields asked for such as documents or fees), and procedures are ranked by inverse document frequency over whole Vietnamese syllables[21], with a penalty for mismatched diacritics, a bonus for adjacent matches, and thresholds on score and coverage (the share of the question's weight found in the name). A policy module checks the plan against the data (real identifiers, commune scope, province versions, at most three procedures) and outputs the behavior. Answers are composed by code from the requested fields of the record, with the source. An optional LLM rewriting step is off in all experiments: we measure the decision, not the wording.

Rule layer (A). Rules written from the base system's errors on the development set, each behind a switch; with all switches off, the system reproduces the base output on every test question.

- *Letter stripping*: remove greetings, the addressed agency, self-introductions and thanks, which otherwise

match procedure names.

- *Citizen lexicon*: about twenty patterns map citizen wording to current official names, including terms changed by recent laws (the household registration book, abolished by the 2020 Law on Residence; “citizen identity card”, renamed “identity card” in 2023).
- *Weak-name filter*: a segment sharing a few words with a very long procedure name does not name it.
- *Out-of-scope gate*: patterns for the status of one's own file, complaints and petitions, law and penalties, police internal matters and recruitment, and topics absent from the data yield *refuse*; explicit procedural questions (how to file a complaint) are exempt.
- *Focus*: in long letters, keep only segments that ask something.
- *Organization default*: without a mentioned organization, prefer the household or individual version.

Second round (A2). Once the first test set had been run and its results read, a second round, behind one more switch, was written on the remaining errors of A on the development and first test sets. It makes the lexicon tolerate inserted words; lets a request recognized by the lexicon override a procedure inferred from a life event in the story (a parent's death suggests death registration, but the question asks to transfer the parent's car); adds out-of-scope patterns (status in plain words, police staff asking about their benefits, private disputes); answers questions on the processing time set by regulation; and answers with the version common to all provinces instead of asking for the province, as the labeling guideline prescribes.

Reranker (B). Candidates are the rule ranker's top eight and the ten in-scope procedures closest to the question under BGE-M3[22]; a cross-encoder, bge-reranker-v2-m3[23], scores each (question, procedure description) pair. The system refuses when the best score is below a threshold and replaces the rule's choice when another candidate wins by a margin; alone, it also answers when the rules find nothing but the best score is positive. Thresholds were chosen by grid search on development (alone: −3.5 and 0.5; on top of A: −8 and 2, no rescue).

LLM reader (C). A local 4-billion-parameter model (Qwen3-4B-Instruct-2507, Q4_K_M quantization, Ollama, temperature 0, JSON schema)[24,25] returns one of six intents (procedure, file status, complaint, law, internal, other) and, for procedure questions, the procedure's name in words, which the rule ranker maps to a procedure; the model never writes an identifier. Non-procedure intents become refusals; the *name-only* variant ignores the intent.

RAG baseline (R). To represent a common design in which the model decides, BGE-M3 retrieves the five in-scope procedures closest to the raw question and the same 4B model returns *answer* with a number, *clarify* or *refuse*; no rule is used. A first, short prompt asked for clarification too often on development; two variants (ordered decision rules, and the same with procedure descriptions) were then declared together before either was run. Their development balanced accuracies were 0.343, 0.445 and 0.432, and the second was run once on each test set. R was added after the first test set had been run, but nothing in it was chosen on a test set.

5. Evaluation Protocol

Order of work. Each test set was sampled, labeled and locked by SHA-256 and a commit; rules and thresholds were developed on the development set only; the code was frozen in a commit; each configuration was run once on the test set and the results were read once. Reusing a test set for selection inflates its estimates[26], so no test set was used to choose anything. A run with all switches off matched the base system on every question. After the first test set had been run, it joined the development set (595 questions) for the second round, which was evaluated on the second test set.

Selection rule. Written before each test run: choose the development configuration with the highest balanced accuracy, subject to end-to-end accuracy not lower than that of the rule layer alone, ties going to end-to-end accuracy. In the first round this selected A + B; A with the name-only reader was excluded by its lower development end-to-end accuracy (65.5% against 69.1%). In the second round, A2 + B tied A2 (balanced accuracy 0.916) with one more end-to-end answer (88 against 87 of 114) and was selected; A2 with the name-only reader (0.925) was excluded (84). These development numbers are optimistic, since A2 was written on the same questions.

Metrics. *Behavior accuracy* is the share of correct three-way decisions. Because most real questions should be refused, a system that always refuses scores 79.2% on the first test set; the main metric is therefore *balanced accuracy*, the mean recall of *answer* and *refuse*[27]; *clarify* questions are too few to weigh equally and are reported separately. *Top-1* is the share of *answer* questions whose correct procedure is ranked first, and *end-to-end* accuracy the share that the system also answers. Proportions carry Wilson 95% intervals[28]; paired comparisons use the exact two-sided McNemar test[29,30].

Setup. All runs use the procedure data of 13 October 2026 on one laptop with an NVIDIA GeForce RTX 4050 (6 GB), paced by a temperature guard that pauses between questions. Frozen commits, the hashes of the question sets and the per-question decisions are in the project repository.

6. Results

6.1 Team-written versus real questions

On the same frozen base system (Table 1), behavior accuracy drops by about 21 points from team-written to real questions, balanced accuracy by about 0.2 and top-1 accuracy by 38 to 59 points. The largest change is in what is asked: 78–79% of real questions should be refused, against 11% of team-written ones, and the base system refuses only about half of them. On the team-written set the rule layer makes no net change in behavior (+4/−4, 400/518 in both cases; correct refusals 52 to 55 of 58), so a team-written set alone would have shown neither the problem nor the fix.

Table 1. Base system (all switches off) on team-written and real questions; Wilson 95% intervals in parentheses.

	Team-written (518)	**Real, dev (282)**	**Real, test (313)**
Behavior accuracy	77.2% (73–81)	56.0% (50–62)	56.5% (51–62)
Balanced accuracy	0.852	0.636	0.647
Top-1 on *answer*	274/381 = 71.9%	7/55 = 12.7%	20/59 = 33.9%
Correct refusals	52/58 = 89.7%	116/220 = 52.7%	132/248 = 53.2%
Share to be refused	11%	78%	79%

6.2 First test set

The rule layer (Table 2) raises balanced accuracy from 0.647 to 0.814 and end-to-end accuracy from 19/59 (95% interval 22–45%) to 35/59 (47–71%); base → A gains 96 and loses 10 questions on behavior ($p \approx 9 \cdot 10^{-19}$) and 22 against 6 end-to-end (p=0.004). Correct refusals rise from 132 to 219 of 248 while correct *answer* decisions stay level (45 to 44). Without the 11 near-duplicates of development questions the picture is the same (302 questions: balanced accuracy 0.651 to 0.810, end-to-end 19 to 34 of 58, +92/−10, $p \approx 9 \cdot 10^{-18}$). Used alone, the reranker (0.692) and the LLM reader (0.785) help less. Added to the rules, the reranker gains one question (+1/−0; the selected configuration, 0.823 and 36/59). The reader's intents add refusals (+25/−10 on behavior against A, p=0.017) but lose answers (32/59; +8/−11 end-to-end, p=0.65); its procedure name alone gives the highest scores (0.834, 39/59), but this configuration was not selected on development and its end-to-end gain over A is not significant (+8/−4, p=0.39). The RAG baseline in which the model decides has the lowest balanced accuracy, 0.489, below always refusing: it asks for clarification on 118 of the 248 questions it should refuse (against A: +16/−141 on behavior, $p \approx 4 \cdot 10^{-26}$; end-to-end 28 against 35, p=0.23). No configuration recognizes more than one of the six *clarify* questions.

Table 2. First locked test set (313 questions: 59 answer, 6 clarify, 248 refuse), one run each. Beh.: behavior accuracy; Bal.: balanced accuracy; Ans., Ref.: correct *answer* and *refuse* decisions; Top-1, Top-3 and E2E (end-to-end) out of 59; for R, Top-3 counts the first three retrieved candidates.

Configuration	Beh. %	Bal.	Ans.	Ref.	Top-1	Top-3	E2E
Always refuse	79.2	0.500	0	248	0	0	0
Base	56.5	0.647	45	132	20	23	19
B: reranker alone	68.7	0.692	40	175	22	30	21
C: LLM reader alone	81.8	0.785	42	213	25	28	25
R: RAG + LLM decides	44.1	0.489	33	104	28	35	28
A: rule layer	84.0	0.814	44	219	37	39	35
A + C (intent and name)	88.8	0.817	40	237	33	33	32
A + C (name only)	84.4	0.834	47	216	41	41	39
A + B (selected)	84.4	0.823	45	219	38	39	36

Rescoring the stored runs on the 91 doubly labeled questions with the second-pass labels gives the same picture: base → A gains 24 and loses 1 on behavior ($p<10^{-5}$; 26/1 with first-pass labels), A and A + B decide identically, and B < C < A is unchanged.

6.3 Second test set

On the new questions (Table 3) the rule layer again changes refusals most: correct refusals rise from 175 to 297 of 335 and balanced accuracy from 0.571 to 0.705 (base → A: +125/−5, $p \approx 4 \cdot 10^{-31}$). The second round does not carry over. A2 + B, which had gained 25 questions on behavior and lost none against A on development, ties A here (+4/−4, p=1). In the comparisons declared before the run, it ties the first-round selection A + B on behavior (+3/−3, p=1; end-to-end 10 against 8 of 21, +3/−1, p=0.63) and beats the base (+129/−9, $p \approx 2 \cdot 10^{-28}$; end-to-end 10 against 5, +7/−2, p=0.18). With 21 answerable questions, end-to-end intervals are wide (A: 9/21, 24–63%; A2 + B: 10/21, 28–68%). R is again last (0.413; +19/−122 on behavior against A2 + B), although it recognizes 4 of the 7 *clarify* questions, against at most one for the other configurations.

Table 3. Second locked test set (363 questions: 21 answer, 7 clarify, 335 refuse), one run each.

Configuration	Beh. %	Bal.	Ans.	Ref.	Top-1	E2E
Always refuse	92.3	0.500	0	335	0	0
Base	52.1	0.571	13	175	5	5
R: RAG + LLM decides	56.8	0.413	5	197	4	4
A: rule layer	85.1	0.705	11	297	9	9
A + B (first-round selection)	85.1	0.683	10	298	8	8
A2 + B (second-round selection)	85.1	0.729	12	297	10	10

7. Discussion

Why rules help. Most of the gain is in refusing correctly: on the first test set, 87 more correct refusals against one fewer correct *answer* decision. Out-of-scope questions in real traffic fall into a few recurring kinds (file status, complaints and petitions, penalties, police internal matters) whose wording is regular enough for patterns and whose cost of error is high. The learned components answer a different question. The reranker scores how well a procedure matches a text, which is high for a complaint that mentions a real procedure; the LLM reader recognizes intents but often labels long narrative procedural questions as complaints or legal questions. Its procedure name alone gave the best first-test scores, but it lowered end-to-end accuracy on development in both rounds, so we do not claim that it helps.

Remaining errors of A. In a post-hoc analysis of the 24 *answer* questions that A misses on the first test set, 12 are refused; for 4 of them the base system had ranked the correct procedure first, so the out-of-scope gate is at fault. Of the 12 that A answers wrongly or clarifies, 7 have the correct procedure outside its top five: long letters in which the procedure is implied rather than named (a car registered to a deceased parent, while the story triggers death registration). A2 was written against such errors.

Why the second round did not carry over. A2 + B changes the decision or the chosen procedure of about as many questions on the new set as on development (5.2% against 6.2%), but, counting a decision as correct only

with the right procedure, the changes were 30 gains and no losses on development and 4 and 4 on the new set. The clearest sign is in refusals: its new out-of-scope patterns turned 15 wrong answers into correct refusals on development but one of the 335 questions to be refused in the new set, which suggests that they were written from particular phrasings. Two other factors may contribute: with 21 answerable questions the new set has little power on the *answer* side, and its labels were proposed by a different model with a slightly extended guideline. Either way, a gain of 25 questions on reused questions became a tie on new ones.

Limitations and threats to validity. (i) One domain, language and data snapshot; the rules are hand-written for Vietnamese commune-level procedures, and the results say nothing about provincial or ministry procedures or a general assistant for all procedures. (ii) The test sets have only 59 and 21 *answer* and 6 and 7 *clarify* questions, so small differences are not significant and nothing can be concluded about clarification. (iii) The development and first test labels were produced with the same AI assistant that wrote the rules; the second-pass sample and the second test set were labeled with another model and checked item by item by an author, who had seen the aggregate class proportions of the first test set. Agreement is substantial and conclusions hold under either label set, but no fully independent human annotation exists. (iv) All questions come from the same two portals, and 79% refusals in our samples understate the share in random traffic. (v) Systems from other studies could not be run on our questions, and their published numbers measure different tasks; our external baseline is a RAG design we implemented. (vi) LLM components were run once at temperature 0 on one quantized model.

8. Conclusion

For an assistant limited to commune-level procedures, the hard part of real citizen questions is not finding a procedure but deciding whether to answer at all. A system that looked strong on team-written questions answered nearly half (114 of 248) of the real questions it should have refused. A small, transparent rule layer nearly doubled the end-to-end correct answers on a locked test set and kept its gain in refusals on a second one, while a reranker and a local 4B LLM reader added no significant end-to-end gain and a RAG design in which the model decides did worse. A second round of rules that looked clearly better on development was no better on new questions: gains must be measured on questions locked before the work that produced them. Future work will let the model propose procedure names while rules keep the decision, collect more *answer* and *clarify* cases, and replace the AI-assisted labels with independent human annotation.

Ethics and Data Availability

The procedure records are public administrative information. The National Public Service Portal has no official API; records were collected once (13 October 2026) through the portal's public, unauthenticated JSON endpoints used by its own web pages, at two requests per second on a single thread, without logging in or bypassing any access control, and contain no personal data. Citizen questions come from two public Hugging Face datasets whose cards state no license; personal data were masked before labeling, no names or identifiers appear in this paper, and no data were collected from human participants, so no ethics approval was required. Because the licenses are unstated, we do not redistribute the question texts. The code, the locked question identifiers and hashes, the labels and the per-question decisions of every run, without question texts, are available in the project repository **[TODO: URL]**; each identifier maps to a row of the original datasets.

Competing Interests

The authors declare no competing interests.

Declaration of AI Use

During the preparation of this work, the authors used Claude Opus 5.5 (Anthropic) to assist with writing and modifying the software, running and summarizing the experiments, annotation of the development set (two independent passes, reconciled and reviewed by the project lead) and the first test set, and drafting and language editing of the manuscript; the code that builds the procedure database was first written with Claude Opus 5 and later modified with Claude Opus 5.5. The same assistant that produced the first-pass labels also wrote the rules evaluated here. Gemini 4 Argon (Google) proposed the labels of the second-pass sample and of the second test set, which an author reviewed item by item. The research questions, the evaluation protocol and the final decisions were made by the authors, who reviewed, verified and edited the output and take full responsibility for the content of the publication.

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ORCID

[TODO: ORCID iDs of all authors.]

References

1. Government of Viet Nam, “National Public Service Portal”, 2026, \urlhttps://dichvucong.gov.vn, procedure records collected 13 October 2026.
2. A. Kaplan, “GovAI-Pipe: A Layered AI Governance Pipeline for Citizen-Facing AI in Turkey's e-Government Gateway”, 2026, arXiv:2606.01417.
3. L. Samson and I. Gornishka and G. L\^o and Y. M. Asano and S. Ghebreab, “From Values to Benchmarks: Evaluating Large Language Models for Governmental Use in Dutch”, 2026, arXiv:2608.09925.
4. B. Wen and J. Yao and S. Feng and C. Xu and Y. Tsvetkov and B. Howe and L. L. Wang, “Know Your Limits: A Survey of Abstention in Large Language Models”, *Transactions of the Association for Computational Linguistics*, vol. 13, pp. 529–556, 2025, doi:10.1162/tacl_a_00754.
5. A. Kamath and R. Jia and P. Liang, “Selective Question Answering under Domain Shift”, *Proc. 58th Annual Meeting of the Association for Computational Linguistics*, pp. 5684–5696, 2020, doi:10.18653/v1/2020.acl-main.503.
6. S. Larson and A. Mahendran and J. J. Peper and C. Clarke and A. Lee and P. Hill and J. K. Kummerfeld and K. Leach and M. A. Laurenzano and L. Tang and J. Mars, “An Evaluation Dataset for Intent Classification and Out-of-Scope Prediction”, *Proc. 2019 Conf. Empirical Methods in Natural Language Processing and 9th Int. Joint Conf. Natural Language Processing (EMNLP-IJCNLP)*, pp. 1311–1316, 2019, doi:10.18653/v1/D19-1131.
7. A. Pham Duy and H. Le Thanh, “A Question-Answering System for Vietnamese Public Administrative Services”, *Proc. 12th Int. Symp. Information and Communication Technology (SoICT 2023)*, 2023, doi:10.1145/3628797.3628965.
8. D. M. Jimenez-Gutierrez and A. Cirillo and R. Nicolussi and A. Beltrame and A. Vitaletti, “GuidaPA: Privacy-Preserving Chatbot for Public Administration via Federated Learning”, 2026, arXiv:2606.01386.
9. M.-T. Nguyen and K.-T. Tran and N. V. Nguyen and X.-S. Vu, “ViGPTQA – State-of-the-Art LLMs for Vietnamese Question Answering: System Overview, Core Models Training, and Evaluations”, *Proc. 2023*

Conf. Empirical Methods in Natural Language Processing: Industry Track, pp. 754–764, 2023, doi:10.18653/v1/2023.emnlp-industry.70.

10. P. T. X. Hien and D. N. T. Nhi and P. T. N. Huyen, “Integrating Information Retrieval and Large Language Models for Vietnamese Legal Document Query Systems”, *Proc. 17th Int. Joint Conf. Knowledge Discovery, Knowledge Engineering and Knowledge Management (IC3K 2025), Vol. 2: KEOD and KMIS*, pp. 290–300, 2025, doi:10.5220/0013751200004000.
11. P. Rajpurkar and R. Jia and P. Liang, “Know What You Don't Know: Unanswerable Questions for SQuAD”, *Proc. 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pp. 784–789, 2018, doi:10.18653/v1/P18-2124.
12. M. Aliannejadi and H. Zamani and F. Crestani and W. B. Croft, “Asking Clarifying Questions in Open-Domain Information-Seeking Conversations”, *Proc. 42nd Int. ACM SIGIR Conf. Research and Development in Information Retrieval*, pp. 475–484, 2019, doi:10.1145/3331184.3331265.
13. P. Lewis and E. Perez and A. Piktus and F. Petroni and V. Karpukhin and N. Goyal and H. K\"uttler and M. Lewis and W.-t. Yih and T. Rockt\"aschel and S. Riedel and D. Kiela, “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks”, *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, pp. 9459–9474, 2020.
14. A. Asai and Z. Wu and Y. Wang and A. Sil and H. Hajishirzi, “Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection”, *Proc. 12th Int. Conf. Learning Representations (ICLR 2024)*, 2024.
15. S.-Q. Yan and J.-C. Gu and Y. Zhu and Z.-H. Ling, “Corrective Retrieval Augmented Generation”, 2024, arXiv:2401.15884.
16. T. Rebedea and R. Dinu and M. N. Sreedhar and C. Parisien and J. Cohen, “NeMo Guardrails: A Toolkit for Controllable and Safe LLM Applications with Programmable Rails”, *Proc. 2023 Conf. Empirical Methods in Natural Language Processing: System Demonstrations*, pp. 431–445, 2023, doi:10.18653/v1/2023.emnlp-demo.40.
17. hieunguyen1053, “dichvucong_faqs (config hoi-dap)”, *Hugging Face dataset*, 2026, \urlhttps://huggingface.co/datasets/hieunguyen1053/dichvucong_faqs, accessed 18 October 2026.
18. doanhieung, “vn-mps-public-service-qa”, *Hugging Face dataset*, 2026, \urlhttps://huggingface.co/datasets/doanhieung/vn-mps-public-service-qa, accessed 18 October 2026.
19. J. Cohen, “A Coefficient of Agreement for Nominal Scales”, *Educational and Psychological Measurement*, vol. 20, pp. 37–46, 1960, doi:10.1177/001316446002000104.
20. J. R. Landis and G. G. Koch, “The Measurement of Observer Agreement for Categorical Data”, *Biometrics*, vol. 33, pp. 159–174, 1977, doi:10.2307/2529310.
21. S. Robertson and H. Zaragoza, “The Probabilistic Relevance Framework: BM25 and Beyond”, *Foundations and Trends in Information Retrieval*, vol. 3, pp. 333–389, 2009, doi:10.1561/1500000019.
22. J. Chen and S. Xiao and P. Zhang and K. Luo and D. Lian and Z. Liu, “M3-Embedding: Multi-Linguality, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation”, *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 2318–2335, 2024, doi:10.18653/v1/2024.findings-acl.137.
23. BAAI, “bge-reranker-v2-m3, model card”, 2024, https://huggingface.co/BAAI/bge-reranker-v2-m3, accessed 25 October 2026.
24. Qwen Team, “Qwen3 Technical Report”, 2025, arXiv:2505.09388.
25. Qwen Team, “Qwen3-4B-Instruct-2507, model card”, 2025, https://huggingface.co/Qwen/Qwen3-4B-Instruct-2507, accessed 25 October 2026.
26. C. Dwork and V. Feldman and M. Hardt and T. Pitassi and O. Reingold and A. Roth, “The Reusable Holdout: Preserving Validity in Adaptive Data Analysis”, *Science*, vol. 349, pp. 636–638, 2015, doi:10.1126/science.aaa9375.

27. K. H. Brodersen and C. S. Ong and K. E. Stephan and J. M. Buhmann, “The Balanced Accuracy and Its Posterior Distribution”, *Proc. 20th Int. Conf. Pattern Recognition (ICPR 2010)*, pp. 3121–3124, 2010, doi:10.1109/ICPR.2010.764.
28. E. B. Wilson, “Probable Inference, the Law of Succession, and Statistical Inference”, *Journal of the American Statistical Association*, vol. 22, pp. 209–212, 1927, doi:10.1080/01621459.1927.10502953.
29. Q. McNemar, “Note on the Sampling Error of the Difference between Correlated Proportions or Percentages”, *Psychometrika*, vol. 12, pp. 153–157, 1947, doi:10.1007/BF02295996.
30. T. G. Dietterich, “Approximate Statistical Tests for Comparing Supervised Classification Learning Algorithms”, *Neural Computation*, vol. 10, pp. 1895–1923, 1998, doi:10.1162/089976698300017197.